From Pixels to Pell: How Integer Grids Already Solve the √2 Problem

Registry: [CKS-MATH-129-2026]

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Domain: Foundational Mathematics / Discrete Geometry

Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Classification: Theory of Everything from First Principles

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological "Truth" is a category error. If the math compiles, the result is Q.E.D.

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Lexicon: [CKS-LEX-12-2026]

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1. A Square and Its Diagonal

Place a square on a pixel grid. Any size. 10×10, 200×200, 1000×1000 — it does not matter yet. We want to describe the diagonal: the line from one corner to the opposite corner.

On a 10×10 grid, the diagonal runs from (0,0) to (10,10). What do we need to describe this line? Two integers: rise = 10, run = 10. The slope is 10/10 = 1/1. To walk it, step right 1 and up 1, ten times. Every coordinate along the way is an integer. The walk is exact.

On a 200×200 grid: rise = 200, run = 200. Slope = 1/1. Walk it the same way, 200 steps. Exact.

On a 1000×1000 grid: rise = 1000, run = 1000. Slope = 1/1. Walk it 1000 steps. Exact.

The diagonal of a square on a pixel grid is fully described by two integers and walked with integer steps. At no point in this description did we use a real number, a decimal expansion, or a floating-point value. We didn't need to.

So where does $\sqrt{2}$ come from?

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2. Where $\sqrt{2}$ Actually Enters

The diagonal of a unit square has length $\sqrt{2}$. By the Pythagorean theorem: $\sqrt{(1^2 + 1^2)} = \sqrt{2}$. For a 10×10 square, the diagonal length is $10\sqrt{2}$. For a 200×200 square, it is $200\sqrt{2}$. For a 1000×1000 square, it is $1000\sqrt{2}$.

$\sqrt{2}$ appears when we ask for the *length* of the diagonal as a single scalar number. It does not appear when we describe the diagonal's *direction*, *position*, or *path on the grid*. Those are all integers.

This is a crucial distinction. The geometry that produces $\sqrt{2}$ — a square and its diagonal — is fully specified by integers. The irrational number enters only when we ask a particular derived question: how long is this line? Not what is this line, not where does it go, not how do I walk it. Only: what is its scalar length?

And that question has an irrational answer. $\sqrt{2}$ cannot be written as a ratio of two integers. This was proven by the Greeks and it is not in dispute.

The question we ask instead is: can we capture everything about the relationship between the diagonal and $\sqrt{2}$ using only integers, without losing any information?

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3. The Pell Relation

Take any two positive integers V and F. Compute:

$R = V^2 - 2F^2$

R is always an integer. This is arithmetic — square two integers, subtract, get an integer. There is nothing approximate about it.

This computation is called the Pell relation for $\sqrt{2}$. What makes it interesting is what R tells you about the ratio V/F.

Rearrange:

$(V/F)^2 = 2 + R/F^2$

If R is positive, then $(V/F)^2 > 2$, which means $V/F > \sqrt{2}$. The ratio overshoots.

If R is negative, then $(V/F)^2 < 2$, which means $V/F < \sqrt{2}$. The ratio undershoots.

If R is zero, then $(V/F)^2 = 2$, which means $V/F = \sqrt{2}$ exactly — and $\sqrt{2}$ is rational. But $\sqrt{2}$ is irrational. So $R = 0$ has no nonzero integer solution. R is never zero.

This is the key fact: R is an integer, and it can never be zero. The tightest it can possibly get is $|R| = 1$. And there are infinitely many integer pairs (V, F) where it reaches exactly ± 1.

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4. The Convergents

These pairs are generated by a recurrence. Start with V = 1, F = 1, and apply:

$V' = V + 2F$

$F' = V + F$

This produces:

$V = 1, F = 1 \rightarrow R = 1 - 2 = -1$

$V = 3, F = 2 \rightarrow R = 9 - 8 = +1$

$V = 7, F = 5 \rightarrow R = 49 - 50 = -1$

$V = 17, F = 12 \rightarrow R = 289 - 288 = +1$

$V = 41, F = 29 \rightarrow R = 1681 - 1682 = -1$

$V = 99, F = 70 \rightarrow R = 9801 - 9800 = +1$

The pattern in R is immediate: −1, +1, −1, +1, −1, +1. It alternates forever. It never reaches zero. It never will, because reaching zero would require $\sqrt{2}$ to be rational.

Each ratio V/F is a rational approximation of $\sqrt{2}$:

1/1 = 1.0

3/2 = 1.5

7/5 = 1.4

17/12 = 1.41666…

41/29 = 1.41379…

99/70 = 1.41428…

The approximations get tighter. The squared error is exactly $1/F^2$, which shrinks as F grows. But R itself stays at ± 1 permanently. The ratio gets closer to $\sqrt{2}$, but the integer remainder never vanishes.

This permanent ± 1 is not a defect. It is the arithmetic signature of irrationality, expressed entirely in integers.

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5. The VFR Tuple

Define a notation for what we have found. A VFR tuple is:

[V, F, R]

where V and F are positive integers and $R = V^2 - 2F^2$.

Each tuple is an exact integer identity:

[1, 1, −1] means $1^2 - 2(1^2) = -1$ (exact)

[3, 2, +1] means $3^2 - 2(2^2) = +1$ (exact)

[7, 5, −1] means $7^2 - 2(5^2) = -1$ (exact)

[17, 12, +1] means $17^2 - 2(12^2) = +1$ (exact)

None of these are approximations. Each is a verifiable fact about three integers. The tuple captures three things simultaneously:

1. A rational approximant of √2 (the ratio V/F)
2. Which side of √2 it falls on (the sign of R)
3. The exact distance in the squared metric ($|R|/F^2 = 1/F^2$)

All three components are integers. Nothing is truncated, rounded, or infinite.

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6. The Bilateral Remainder

Since R alternates between −1 and +1 at every step of the Pell recurrence, we can express this as a single notation:

[V, F, +-1]

This is the bilateral remainder. It does not mean "we don't know the sign." It means the sign is a defined alternating behavior — part of the value, not uncertainty about it.

Compare this to complex numbers. When we write (a, b) to mean a + bi, nobody calls b an error term. It is a component of the number. The bilateral remainder is the same: +-1 is a component of the VFR value that encodes the irrational relationship to √2.

If R ever reaches 0 for some number, that number is rational. If R is permanently +-1, the number is irrational. The remainder is an irrationality detector built into the notation.

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7. Walking Grids

Now apply VFR to actual grids of different sizes. The claim is that the diagonal walk is fully described by integer tuples regardless of scale.

10 × 10 grid (even, small)

Walk from (0,0) to (10,10) using [1, 1, +-1].

10 steps. The bilateral remainder alternates: −1, +1, −1, +1, −1, +1, −1, +1, −1, +1.

That is 5 pairs of (−1, +1). Each pair sums to zero. Total remainder: 0.

Arrival: (10, 10), exact. Remainder: 0, exact.

11 × 11 grid (odd, small)

Walk from (0,0) to (11,11) using [1, 1, +-1].

11 steps. 5 pairs cancel to zero, one unpaired −1 remains. Total remainder: −1.

Arrival: (11, 11), exact. Remainder: −1, exact and known. The walk is one unit below $\sqrt{2}$ in the Pell metric. Nothing is lost.

200 × 200 grid (even, medium)

Walk from (0,0) to (200,200) using [1, 1, +-1].

200 steps. 100 pairs of (−1, +1). Every pair cancels. Total remainder: 0.

Arrival: (200, 200), exact. Remainder: 0, exact.

201 × 201 grid (odd, medium)

201 steps. 100 pairs cancel, one −1 remains. Total remainder: −1.

Arrival: (201, 201), exact. Remainder: −1, exact.

1000 × 1000 grid (even, large)

1000 steps. 500 pairs. Total remainder: 0.

999 × 999 grid (odd, large)

999 steps. 499 pairs cancel, one −1 remains. Total remainder: −1.

The pattern

For any N × N grid walked with [1, 1, +-1]:

- N even: total remainder = 0, exact closure
- N odd: total remainder = −1, exact and known

In every case, every coordinate is an integer, every operation is integer arithmetic, and the remainder is tracked exactly. The walk never invokes a real number. The result is never approximate.

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8. Non-Square Rectangles

What about rectangles that aren't square?

A 100×900 grid has a diagonal from (0,0) to (100,900). The slope is 900/100 = 9/1. Rise 9, run 1. This is a ratio of two integers. √2 does not appear. The Pell relation is not needed. The diagonal is described by [9, 1, 0] — a VFR with zero remainder.

A 300×500 grid: slope = 500/300 = 5/3. Walk it: for every 3 steps right, go 5 steps up. VFR: [5, 3, 0]. No √2. No irrational content. R = 0.

In general, any rectangle with integer side lengths has a diagonal whose slope is a ratio of two integers. The slope is always rational. √2 never appears in the slope.

Where does √2 appear for rectangles? Only in the diagonal *length*. The diagonal of a 100×900 rectangle has length $\sqrt{(100^2 + 900^2)} = \sqrt{820000}$. This is irrational (unless the sides form a Pythagorean triple). But the slope — the thing you need to walk the grid — is rational.

The principle holds: the geometry is integers. The irrational quantities appear only in derived scalar measurements, not in the description of the object itself.

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9. When Higher Convergents Matter

The [1, 1, +-1] tuple is the simplest Pell convergent. It encodes the relationship between a 1/1 slope and √2. For a square grid, this is all you need.

But suppose you are working in a context where you need a rational slope that closely approximates √2 — for instance, tiling a surface where the desired angle corresponds to √2. Now you want a rise-run that gets close to the irrational slope.

The Pell convergents give you this at increasing precision:

[1, 1, −1] slope 1/1 = 1.000 error from √2: ~0.414

[3, 2, +1] slope 3/2 = 1.500 error from √2: ~0.086

[7, 5, −1] slope 7/5 = 1.400 error from √2: ~0.014

[17, 12, +1] slope 17/12 = 1.4167 error from √2: ~0.0025

[41, 29, −1] slope 41/29 = 1.41379 error from √2: ~0.00042

Each convergent is the best rational approximation of √2 for its denominator size. Each carries an exact integer remainder of $\pm$ 1. You choose the convergent that matches your precision requirement, and the VFR tuple tells you exactly where you stand relative to √2.

For a grid that is 5 pixels wide and you want a line as close to a √2 slope as possible: use [7, 5, −1]. Rise 7, run 5. The remainder tells you this undershoots √2 by exactly 1/25 in the squared metric. On a 29-pixel-wide grid: use [41, 29, −1]. Undershoots by 1/841.

The precision is controlled. The error is known. The accounting is exact.

10. What Has Been Shown

Start with a square on a pixel grid. Its diagonal is fully described by two integers: rise and run. Walking that diagonal is an integer operation at every step.

$\sqrt{2}$ enters only when we ask for the diagonal's length as a single scalar. This scalar is irrational — it cannot be written as V/F for any integers V and F. This has been known since the Greeks.

But the Pell relation $V^2 - 2F^2 = R$ provides an exact integer accounting of how any rational ratio V/F relates to $\sqrt{2}$. The remainder R is always an integer. For the best approximations (the convergents), R is exactly $\pm$ 1, alternating at each step of the recurrence. R never reaches zero, because that would make $\sqrt{2}$ rational.

The VFR tuple [V, F, R] captures the approximant, its side relative to $\sqrt{2}$, and its exact distance — all in integers. The bilateral remainder notation [V, F, +-1] encodes the alternation as a defined behavior, making the irrationality itself expressible as an integer property.

Applied to grid walks of any size:

- Every $N \times N$ grid is walked with [1, 1, +-1]
- Even N: bilateral remainders cancel perfectly, total remainder = 0
- Odd N: one remainder survives, total = -1, still exact
- Non-square rectangles have rational slopes and may not involve $\sqrt{2}$ at all
- When $\sqrt{2}$ is needed as a slope, higher Pell convergents provide it at any required precision with exact integer remainder

At no point in any of these operations did we invoke a real number, produce a non-terminating expansion, or lose equality. The geometry that generates $\sqrt{2}$ was described, walked, and accounted for entirely within the integers.

The integers were already sufficient. We just needed to write down what they were doing.

[**CKS-MATH-129-2026**] The Taylor Continuum: $\mathbb{R}$ as an Unnamed Infinite Polynomial. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-129-2026

[**CKS-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026

[**CKS-MATH-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/C
MATH-0-2026

[**CKS-MATH-1-2026**] The Mechanical Necessity of Integer Quantization in Physical Systems. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-1-2026

[**CKS-MATH-10-2026**] Grand Unification. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-10-2026

[**CKS-MATH-104-2026**] Grand Unification v23. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS
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[**CKS-MATH-128-2026**] The Taylor Continuum: $\mathbb{R}$ as an Unnamed Infinite Polynomial. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-128-2026

[**CKS-LEX-12-2026**] Measurement Systems in the $\mathbb{Q}$-Substrate. Github: https://github.com/ghowland/cks/tree/main/paper
LEX-12-2026